D TRACK

“Track the unseen. Stop the chain.”

PROBLEM STATEMENT

- Drug trafficking has gone **Digital** & **Invisible —** and our detection tools are a decade behind
- Traffickers **don't say illegal words** anymore.

Details:

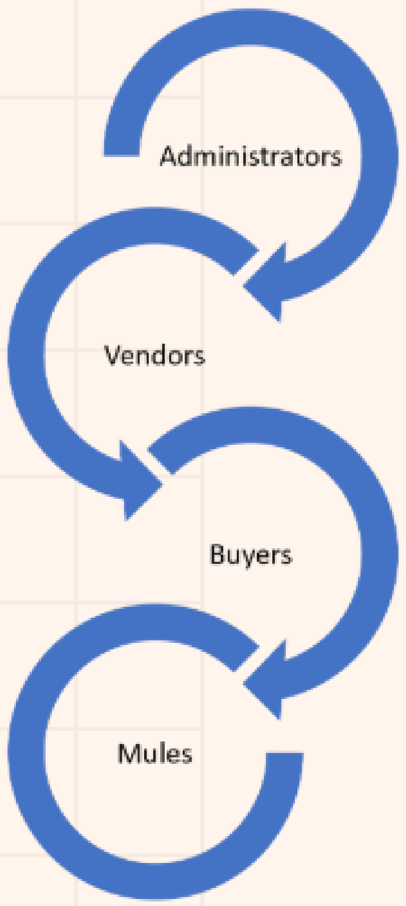
In its 2025 Annual Report, the International Narcotics Control Board:

- India recorded over **1 lakh** drug-related cases in 2025 alone — and that's only what was detected
- **Telegram** alone hosts thousands of active **drug distribution channels** — many operating openly with emoji-coded menus
- Blockchain analytics firm Chainalysis estimates **$24.2 billion in crypto** was received by illicit addresses in 2023

The Silo Problem: Drug networks no longer operate in alleys. They advertise on **Instagram** using coded emoji, transact via **Telegram bots,** and collect payment in **crypto** across three platforms, three identities, zero obvious connection.

The Language Problem: Traffickers use coded language (emojis, slang, patterns) instead of keywords, bypassing **traditional detection systems.**

The Scale Problem: Investigators face massive data with limited resources; without automation, major networks remain hidden while only small cases get caught.



Online Drug Trafficking Network

PROPOSED SOLUTION

AI-Powered Trafficking Detection System

CORE FEATURES:

Slang-to-Signal Detection Engine

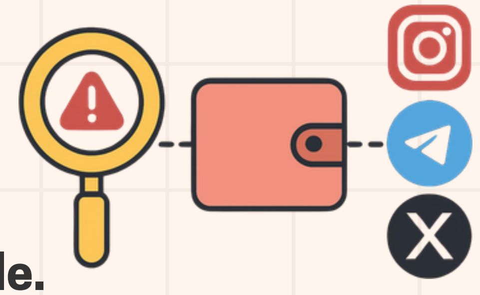
- **Problem it solves**: Traffickers don't say illegal words anymore
- Instead of looking for specific words, **this engine looks for patterns of intent**. It reads a post the way a trained investigator would noticing that
- It **converts every post into a mathematical "meaning vector" using a machine learning model** (sentence-transformers). It then compares that vector against a reference database of confirmed trafficking patterns. If the meaning is close enough flags it

Cross-Platform Handle Stitching

- **Problem it solves**: Traffickers use different platforms (e.g., clean Instagram, active Telegram), making identities hard to link.
- **Uses crypto wallet address as a unique fingerprint** (public, consistent across platforms)

Matches profiles across Instagram, Telegram, and X using
- Same wallet address
- Phone number / bio links
- Similar username patterns

Links **matched accounts into a single unified identity profile.**



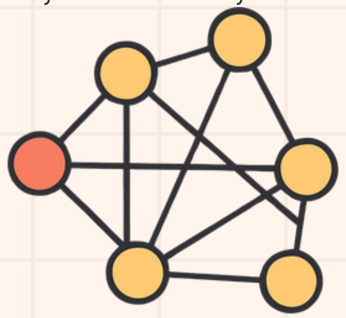
Shadow-Graph Visualizer

- **Problem it solves**: Linked accounts alone don't reveal network structure or key players.

Stores identities and wallets in a graph database :
- Nodes = accounts
- Edges = shared wallet, links, or phone

Renders an interactive network map to explore connections, clusters, and money flows.

Reveals:
- Network scale (isolated vs organized)
- Central wallets (collection points)
- "Clean" accounts promoting operations

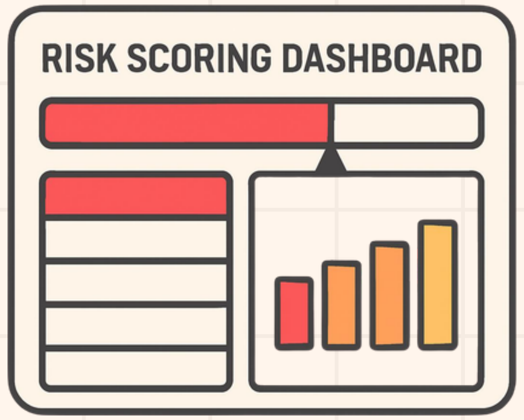


Risk Scoring Dashboard

- **Problem it solves**: Investigators **can't manually review** hundreds of flagged accounts.

Assigns each identity a Risk Score (0–100) based on:
- Graph Centrality: role in network (hub vs peripheral)
- Blockchain Volume: crypto transaction activity
- Activity Frequency: posting/operational intensity

Ranks accounts so high-risk targets appear first.



TECHNICAL ARCHITECTURE

LAYER BASED APPROACH

DATA INGESTION

Instagram / X Scraper
Public posts ,bios ,geostag

Using: Playright , Apify

Telegram Collector
Channel ,bots , media

Using: Telethon

BlockChain API
Wallet history , Volumes

Using: Ethescan , Solscan

PROCESSING

Slang-to-Signal Engine
Converts every scraped post into a semantic embedding vector

OCR Image Pipeline
Extract text from images

Geo Extractor
Location mentions, exif tags

IDENTITY RESOLUTION

Handle Stiching Engine
Public posts ,bios ,geostag

Using: Playright , Apify

Entity Deduplication
Merge duplicate identities

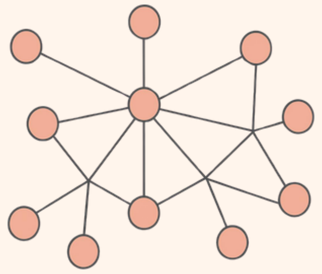
Using: MinHash LSH

SCORING & VISUALISATION

Risk Scoring engine
0–100 per identity, multi-signal

Using: Python · Chain APIs

Graph Database
Nodes,edges, centrality scoring
Using: Neo4j GDS

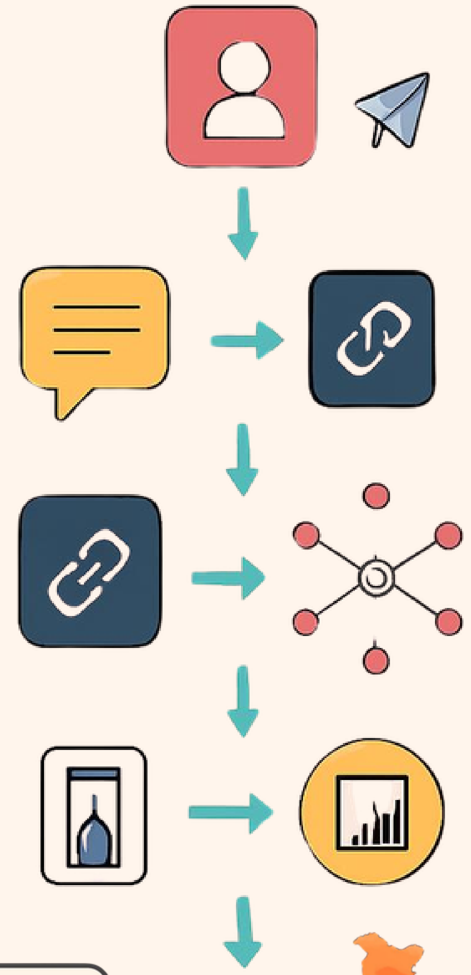


Shadow-Graph Visualizer
Interactive network map, exportable

Risk Dashboard
Ranked targets, filterable

Geospatial Heatmap
Hot-zone mapping, clusters

DATA FLOW



TECH STACK:

PostgreSQL PyTorch neo4j

WHY D-TRACK WINS: THE MVPs

Intent detection, not keyword matching

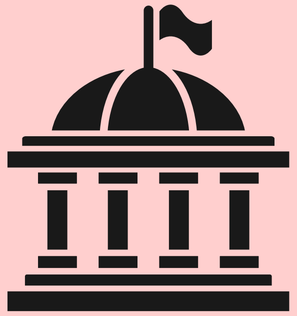
Most tools search for "cocaine" or "weed." TraceNet uses semantic embeddings to detect deal intent — price patterns, emoji proximity, contact handles — even when zero illegal words appear. Traffickers adapted to keyword filters years ago.

Prosecutable graph output, not a flagged list

The output is an interactive network graph showing exactly how a Telegram bot, its Instagram promoters, and wallets are connected. This is evidence-grade intelligence, not a list of suspicious posts. Investigators see a pattern they can act on — not raw data they still need to interpret.

Platform-agnostic by design

The wallet-stitching logic doesn't care what platform it runs on. When traffickers migrate from Telegram to WhatsApp or TikTok, TraceNet adapts — because the crypto wallet always moves with them.



Built to plug into real enforcement pipelines

Designed as an analyst-assist layer — not a black-box decision system — making it compatible with CCTNS (India) and INTERPOL I-24/7 workflows. This is a working deployment path, not a vague "scalability" claim.

IMPACT & BENEFITS

10× Faster Investigations:
Reduces weeks of manual cross-referencing to under 10 minutes — investigators spend time acting on leads, not chasing them.

Cross-Platform Identity Resolution:
Stitches aliases across Instagram, Telegram, and X using shared wallet addresses and bio-link patterns into a single unified profile.

The Anonymity Gap — Closed:
Crypto wallet acts as a permanent fingerprint. Even with aliases and coded language, D-TRACK links the "clean" persona to the distribution network.

Scalable & Broadly Applicable:
Platform-agnostic architecture extends to TikTok, WhatsApp, and beyond. Same logic applies to counterfeit goods, financial fraud, and human trafficking networks.

FROM SCATTERED SIGNALS TO PROSECUTABLE INTELLIGENCE

Metric	Impact
Investigation Time	Reduced from **weeks → 10 minutes** of automated graph traversal
Identity Resolution	**3+ platforms** unified under one wallet fingerprint
Deployment Reach	Integrates with **CCTNS** & **INTERPOL I-24/7** workflows
Data Privacy	100% public data — **no private surveillance** involved